ADVANCE DIGITAL IMAGE PROCESSING
LECTURE # 1

INTRODUCTION

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Course Contents

- **Objective**
  - **Study fundamentals and advanced concepts of Digital Image Processing**
  - **Carry out programming on image processing related issues and obtain publishable results**
- **Textbook**
  - **R. C. Gonzalez and R. E. Woods, *Digital Image, processing*, 2nd or 3rd edition, Prentice Hall**
- **Reference**
  - ***Anil K. Jain*, FUNDAMENTALS OF DIGITAL IMAGE PROCESSING, *Prentice Hall***
  - **Rafael C. Gonzalez and Richard E. Woods, "DIGITAL IMAGE PROCESSING USING MATLAB", PEARSON EDUCATION, 2004.**

Course Outline

- Introduction to Digital Image Processing
- Applications of Digital Image Processing
- Digital Image Fundamentals
- Image enhancement in spatial domain
- Image enhancement in frequency domain
- Color Image Processing
- Image Compression
- Morphological Image Processing
- Real time Applications and Problems in DIP

Digital Image Processing Motivation

- Medical field
- Remote sensing
- Machine/Robot vision
- Forensic investigation
- Smart City/Car/…
- Many more

Image

"A picture is worth a thousand words"

Anonymous

What is an Image?

- Image is a source of information according to information theory

- An image may be defined as a two dimensional function f(x,y) where x and y are spatial coordinates and amplitude of f at any pair of coordinates(x,y) is called the intensity or Gray level of the image at that point.

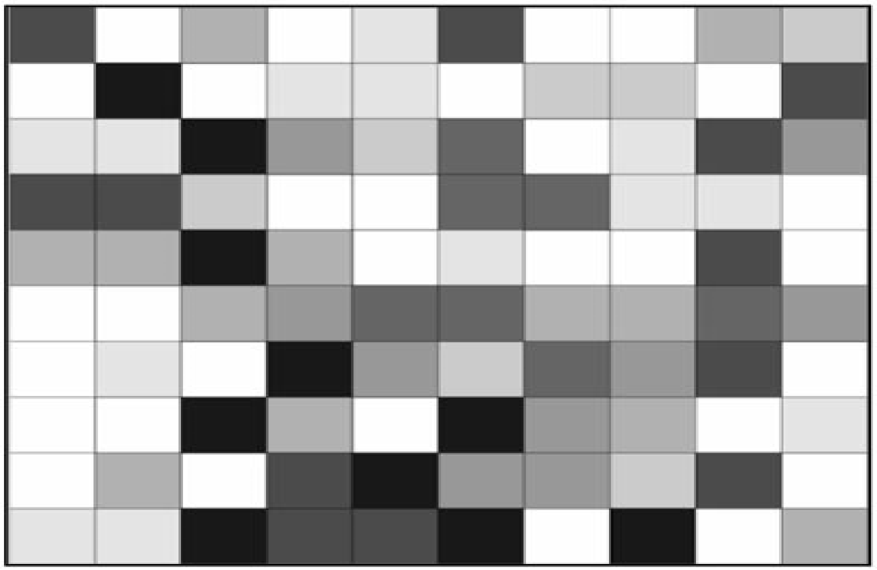
Digital Image

- When x,y and the amplitude values of f are all finite, discrete quantities, we call the image a Digital Image.
- A digital Image is composed of a finite number of elements each of which has a particular location and value
- These elements are referred to as Picture Elements, Image Elements, Pels or Pixels.
- In digital imaging, a pixel is the smallest piece of information in an image.

Pixel

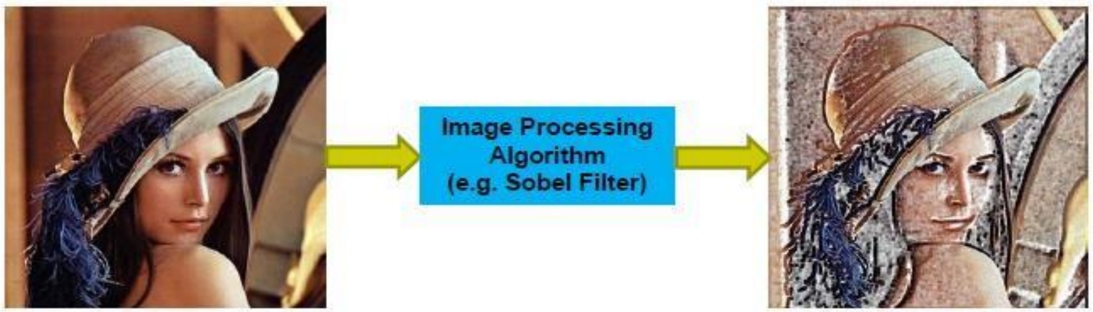
- Pixels are normally arranged in a regular 2-dimensional grid, and are often represented using dots or squares

- The intensity of each pixel is variable; in grayscale images we have one color value while in color systems, each pixel has typically three or four components such as red, green, and blue, or cyan, magenta, yellow, and black



Digital Image Processing

- Image Processing deals with algorithms that transform an input image into a new image (processed image)
- DIP is the field of processing digital images by means of a digital computer

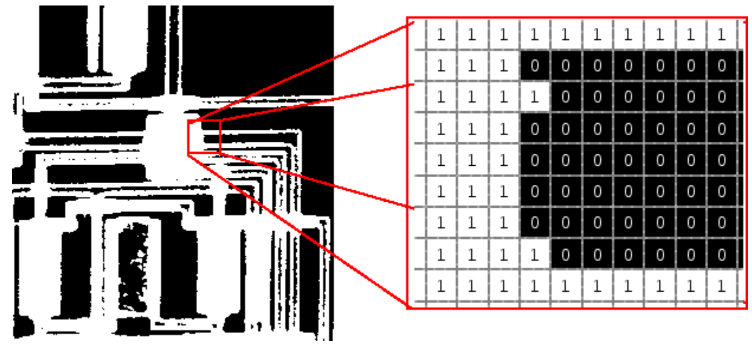


Original Image *Processed Image*

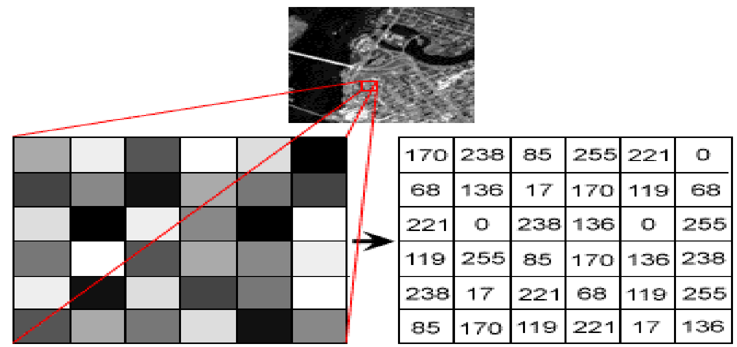
- Digital image processing focuses on two major tasks
  - ✓ Improvement of pictorial information for human interpretation
  - ✓ Processing of image data for storage, transmission and representation for autonomous machine perception

Image Types

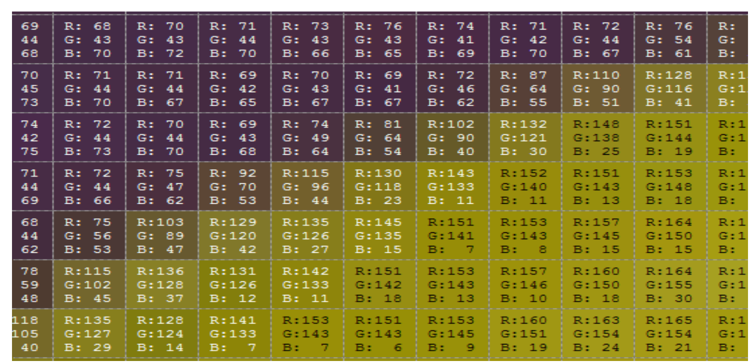
- Binary Image
  - 1 Sample per point



- Gray scale Image
  - 1 Sample per point

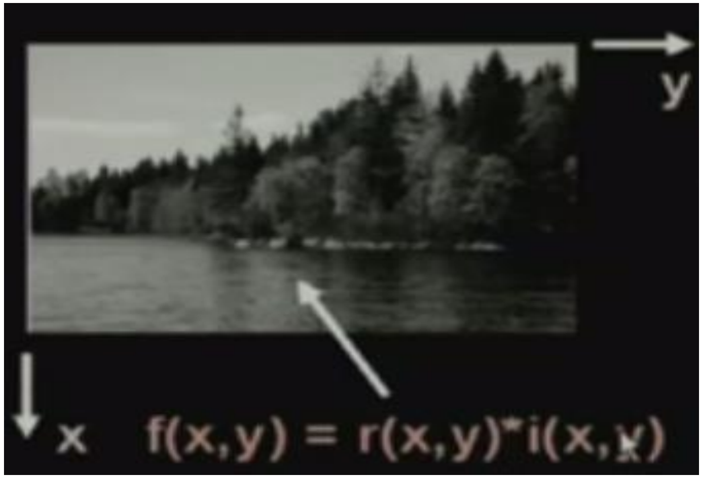


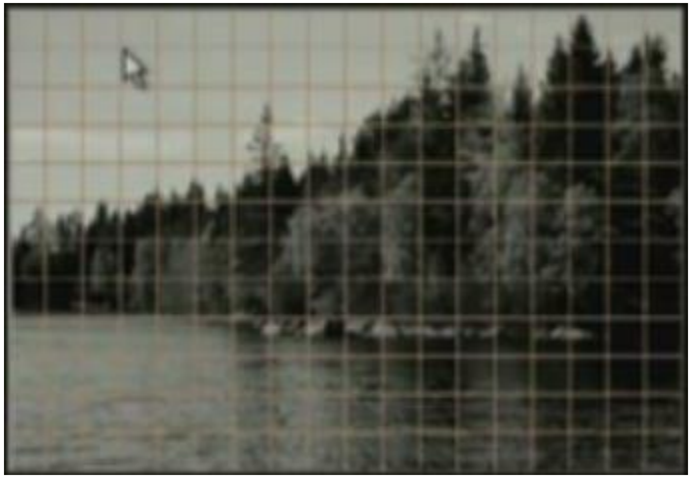
- Color Image
  - 3 or 4 Samples per point



Digital Image Representation

- A digital image f(x,y) is discretized both in spatial coordinates and brightness
- It can be considered as a matrix whose row column indices specify a point in the image and the element value identifies the gray level value at that point
- Spatial discretization by Sampling
- Intensity discretization by quantization





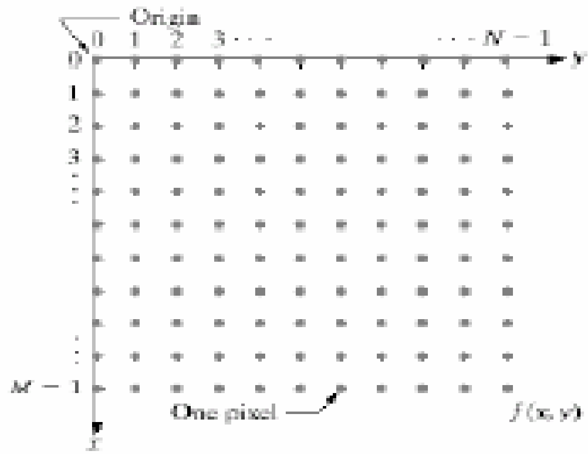
Digital Image Representation

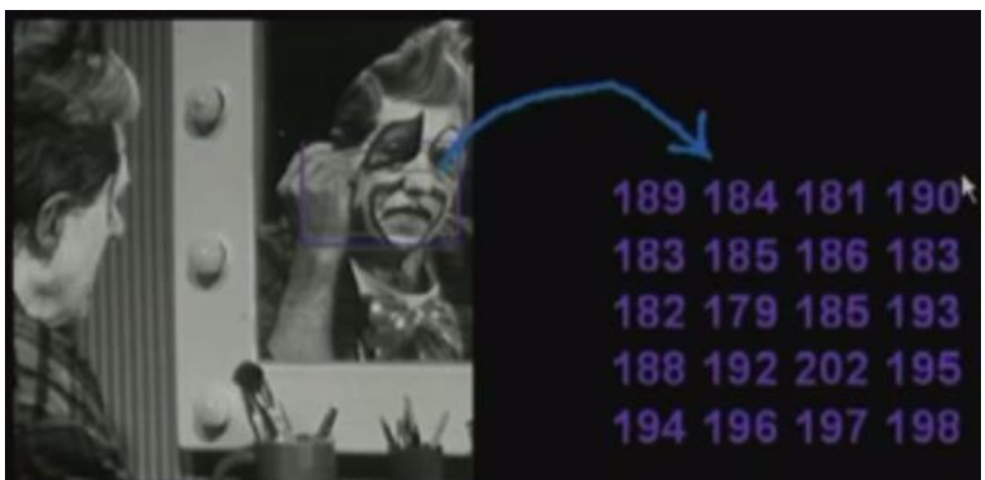
Digital image is expressed as

$$
\begin{array}{c}
\xrightarrow{\text{Columns}} \\
\text{Rows}\downarrow
\begin{bmatrix}
f(0,0) & f(0,1) & \cdots & f(0,N-1) \\
f(1,0) & f(1,1) & \cdots & f(1,N-1) \\
\vdots & \vdots & & \vdots \\
f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1)
\end{bmatrix}
\end{array}
$$

N : No of Columns

M : No of Rows





Digital Image Representation in MATLAB

- The representation of an M×N numerical array in MATLAB

$$f(x,y) = \begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \dots & \dots & \dots & \dots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$

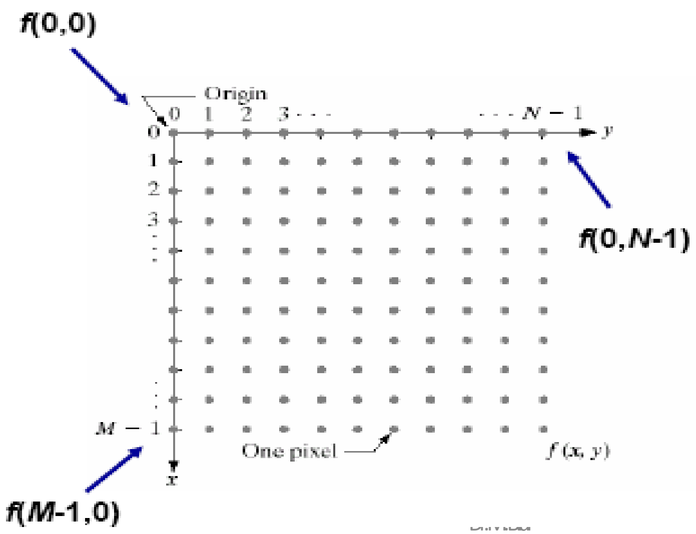


FIGURE 2.18 Coordinate convention used in this book to represent digital images.

L: # of discrete gray levels

$L = 2^k$

$b = M \times N \times k$

Video [8]

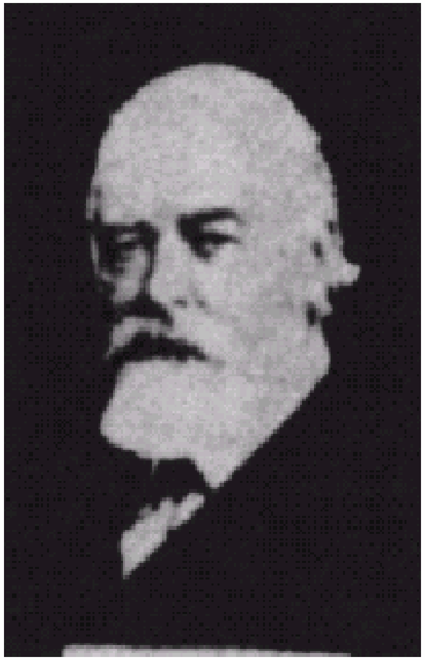
- A digital video consists of **frames** that are presented to the viewer's eye in rapid succession to create the impression of movement. "Digital" in this context means both that a single frame consists of pixels and the data is present as binary data, such that it can be processed with a computer. Each frame within a digital video can be uniquely identified by its **frame index**, a serial number.

A Historical Overview of DIP

- **Bartlane cable picture transmission system** was a technique invented in 1920 to transmit images over cable lines between London and New York in 1920s. It was named after its inventors Harry G. Bartholomew and Maynard D. McFarlane and was first used to transmit a picture across the Atlantic in 1920. Using the Bartlane system, images could be transmitted across the Atlantic in less than three hours. [2]

A Historical Overview of DIP

- The number of distinct gray levels coded by Bartlane system was improved from 5 to 15 by the end of 1920s.[2]



The Boom of Digital Images in the Last 20 Years

- Acquisition
  - Digital cameras, scanners
  - Infrared and microwave imaging etc
- Transmission
  - Internet, satellite and wireless communication
- Storage
  - CD/DVD, Blu-ray
  - Flash memory
- Display
  - CRT monitors, LCD monitor, LED Monitors
  - PDAs, smart phones, smart watches

Key Stages in DIP

- Image Acquisition
- Image Enhancement
- Image Restoration
- Image Compression
- Color Image Processing
- Morphological Image Processing
- Image Segmentation
- Representation and Description
- Image Recognition

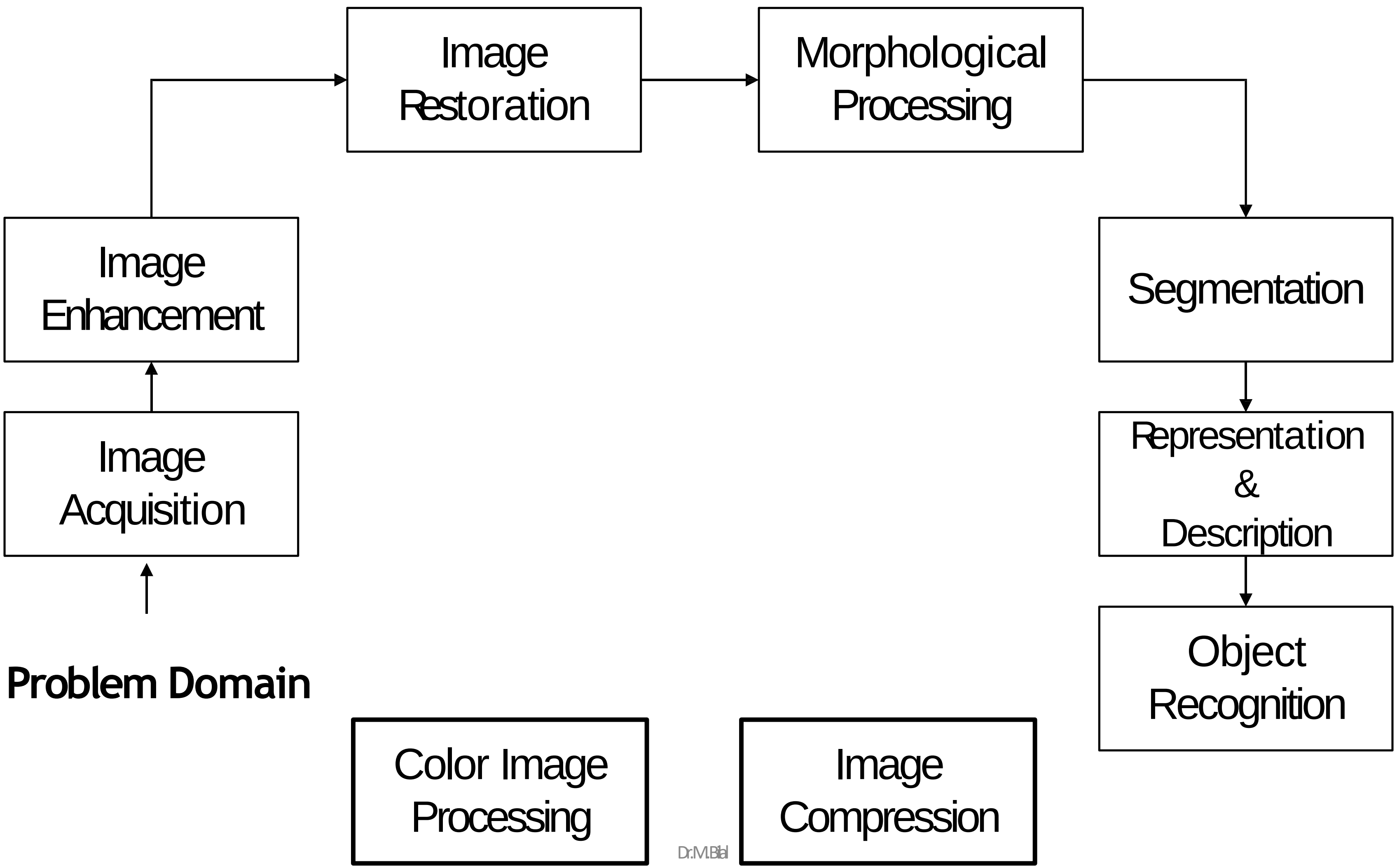
Classification of DIP and Computer Vision Processes

- Image Processing *image in -> image out*
- Image Analysis *image in -> measurements out*
- Image Understanding *image in -> high-level description out*

These are somewhat artificial boundary

- Low-level process: (DIP)
    - Primitive operations where inputs and outputs are images Major functions: image pre-processing like noise reduction, contrast enhancement, image sharpening, etc.
- Mid-level process (DIP and Computer Vision and Pattern Recognition)
    - Inputs are images, outputs are attributes (e.g., edges). major functions: segmentation, description, classification / recognition of objects
- High-level process (Computer Vision)
    - Make sense of an ensemble of recognized objects; perform the cognitive functions normally associated with vision

Key Stages in Digital Image Processing



Key Stages in Digital Image Processing

Image Restoration

Morphological Processing

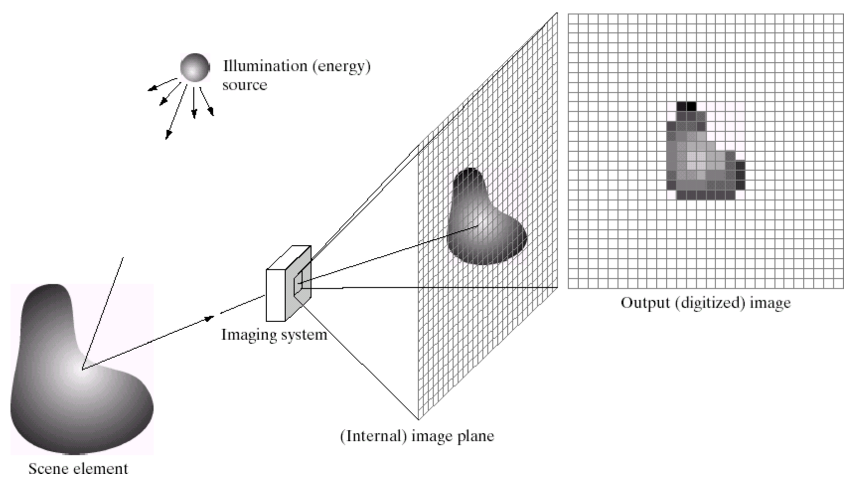
Image Enhancement

Segmentation

Image Acquisition

Representation & Description

Problem Domain



Object Recognition

Color Image Processing

Image Compression

Image Acquisition

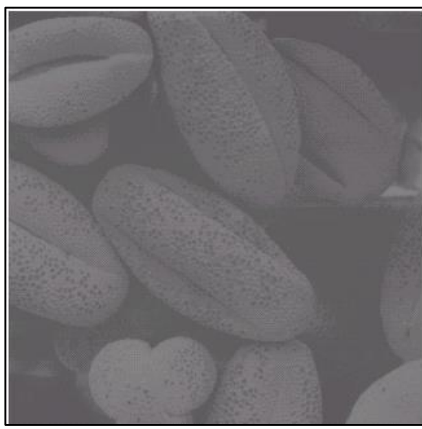
- The first stage of any vision system is the image acquisition stage.
- An image is captured by a sensor (such as a monochrome or color TV camera) & digitized
- If the output of the camera or sensor is not already in digital form, an ADC converter digitizes it
- Images are processed after acquisition.
- However, if the image has not been acquired satisfactorily then the intended tasks may not be achievable

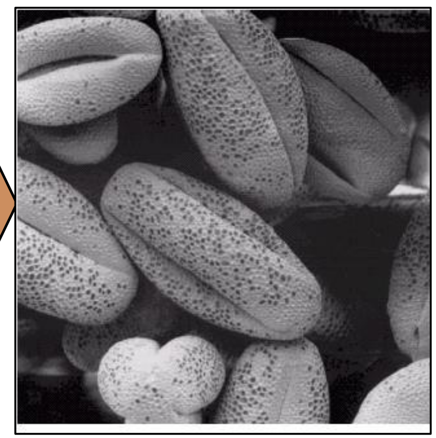
Key Stages in Digital Image Processing

Image Restoration

Morphological Processing

Image Enhancement





Segmentation

Image Acquisition

Representation & Description

Problem Domain

Object Recognition

Color Image Processing

Image Compression

Image Enhancement

- The aim of image enhancement is to improve the perception of information in images for human viewers, or to provide better input for other automated image processing techniques.

- Image enhancement techniques can be divided into two broad categories:

  - ✓ Spatial domain methods, which operate directly on pixels, and
  - ✓ Frequency domain methods, which operate on the Fourier transform of an image.

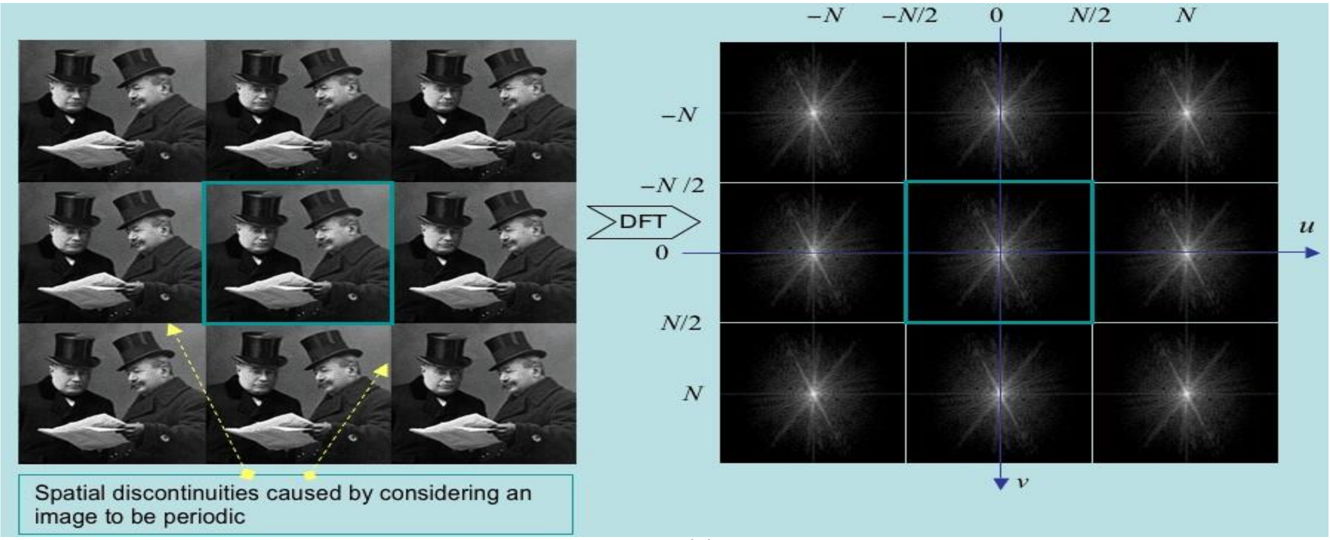
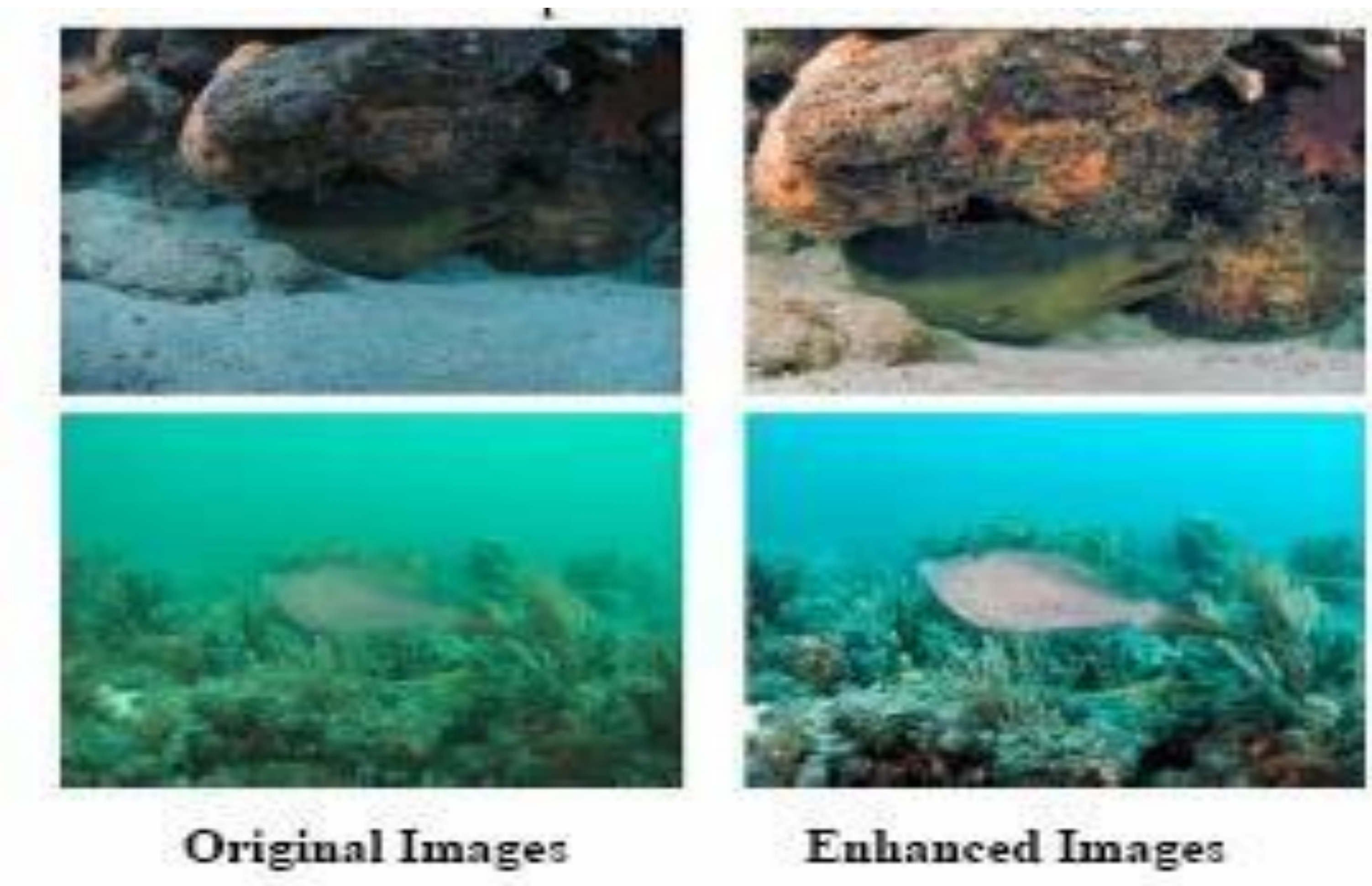


Image Enhancement

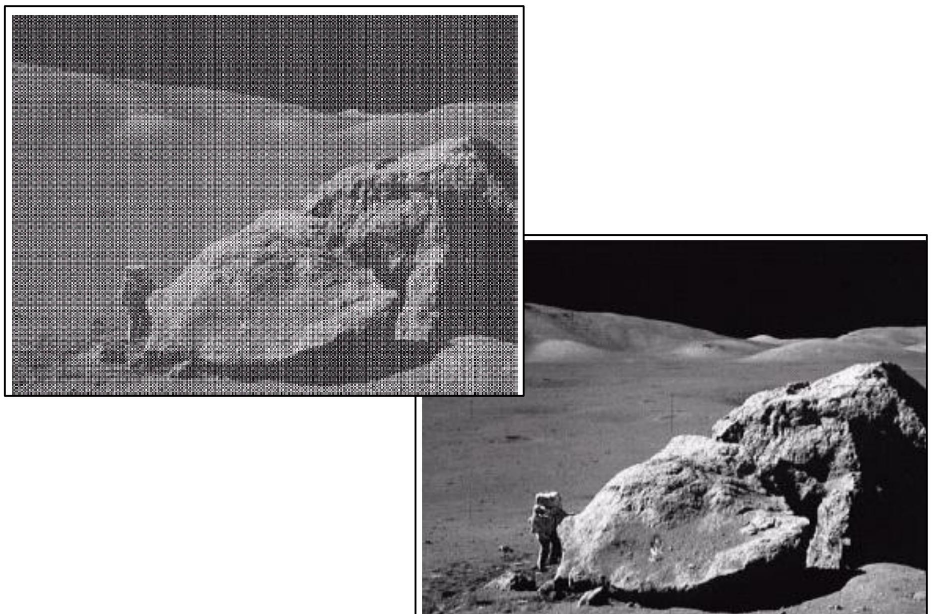


Key Stages in Digital Image Processing

Image Restoration

Morphological Processing

Image Enhancement



Segmentation

Image Acquisition

Representation & Description

Problem Domain

Object Recognition

Color Image Processing

Image Compression

Image Restoration

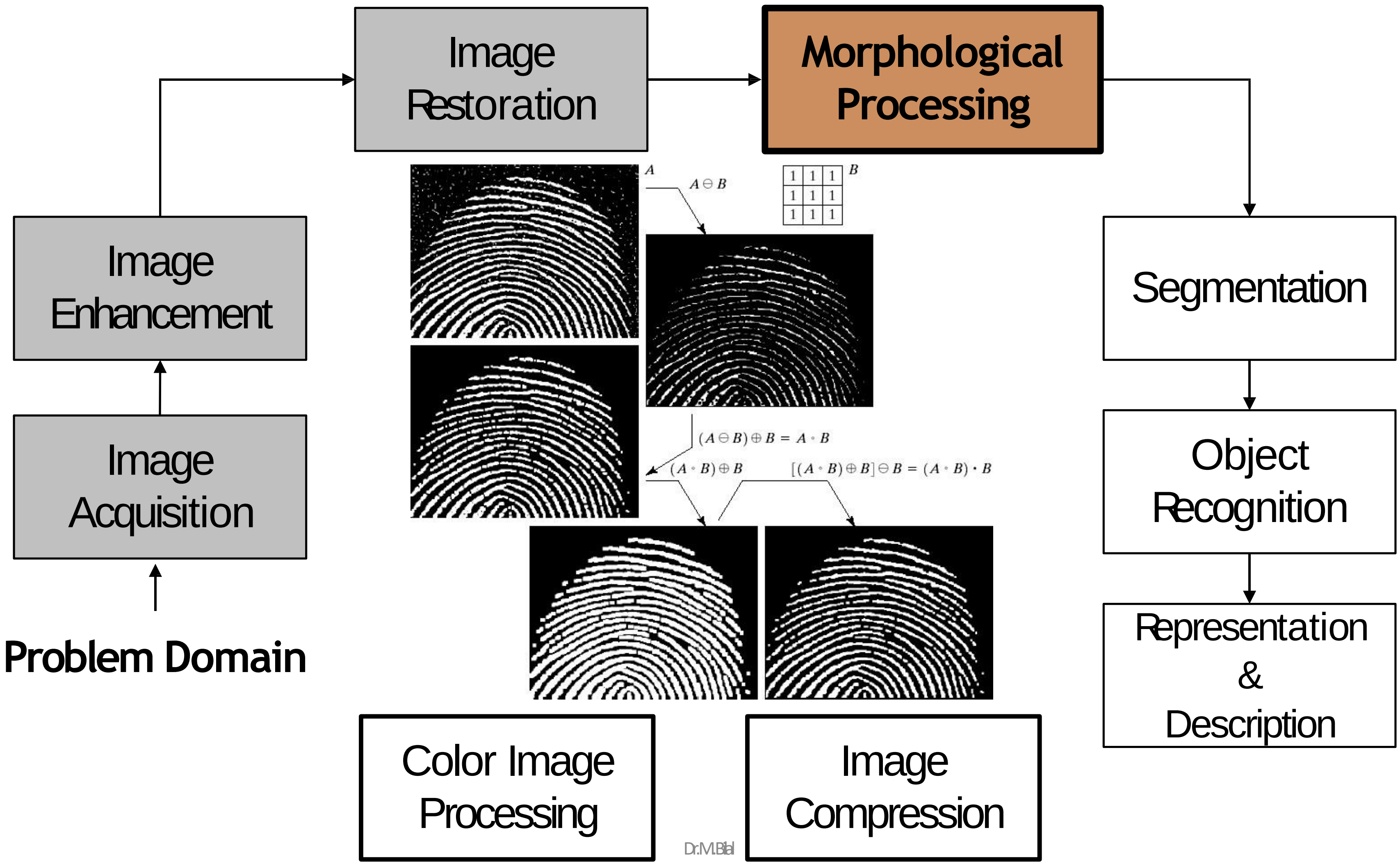
- Image restoration refers to the recovery of an original signal from degraded observations.

- The purpose of image restoration is to "compensate for" or "undo" defects which degrade an image. Degradation comes in many forms such as motion blur, noise, and camera misfocus.

Damaged Image

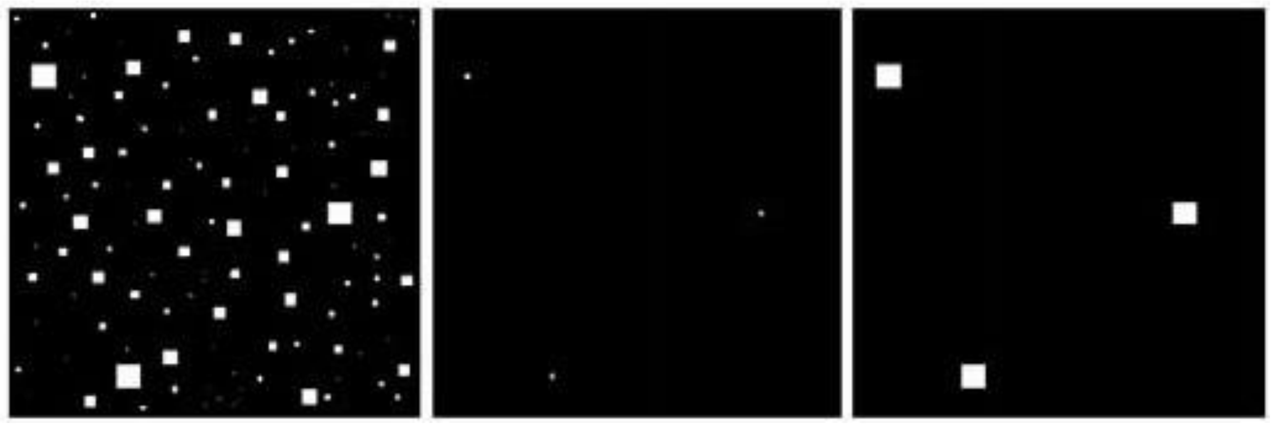
Restored Image

Key Stages in Digital Image Processing



Morphological Image Processing

- Deals with Tools for extracting image components that are useful in the representation & description of shape



Key Stages in Digital Image Processing

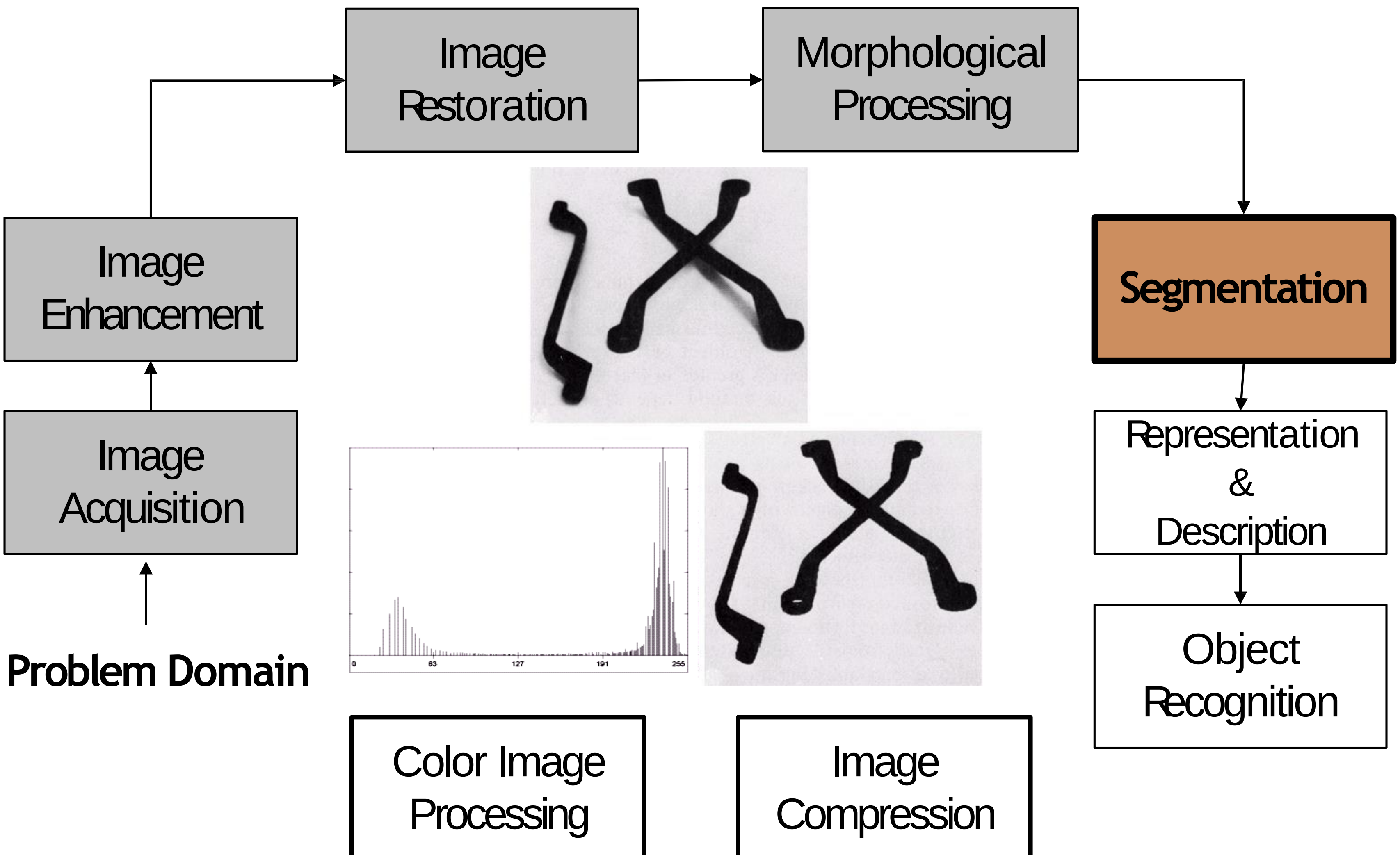
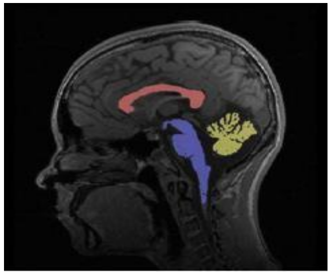


Image Segmentation



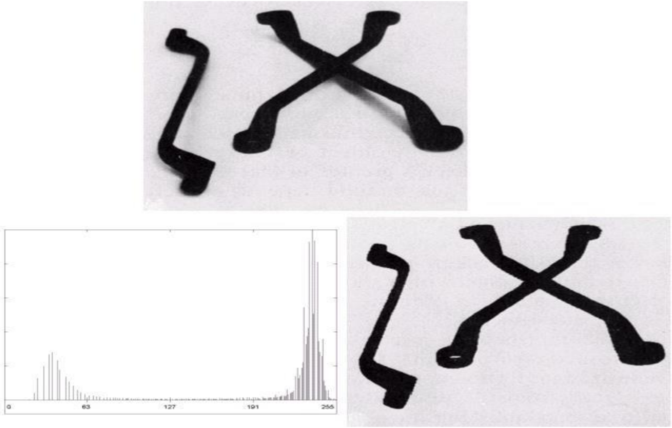
- Segmentation refers to the process of partitioning a digital image into multiple segments (sets of pixels, also known as super pixels). The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze
- Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images

Image Segmentation

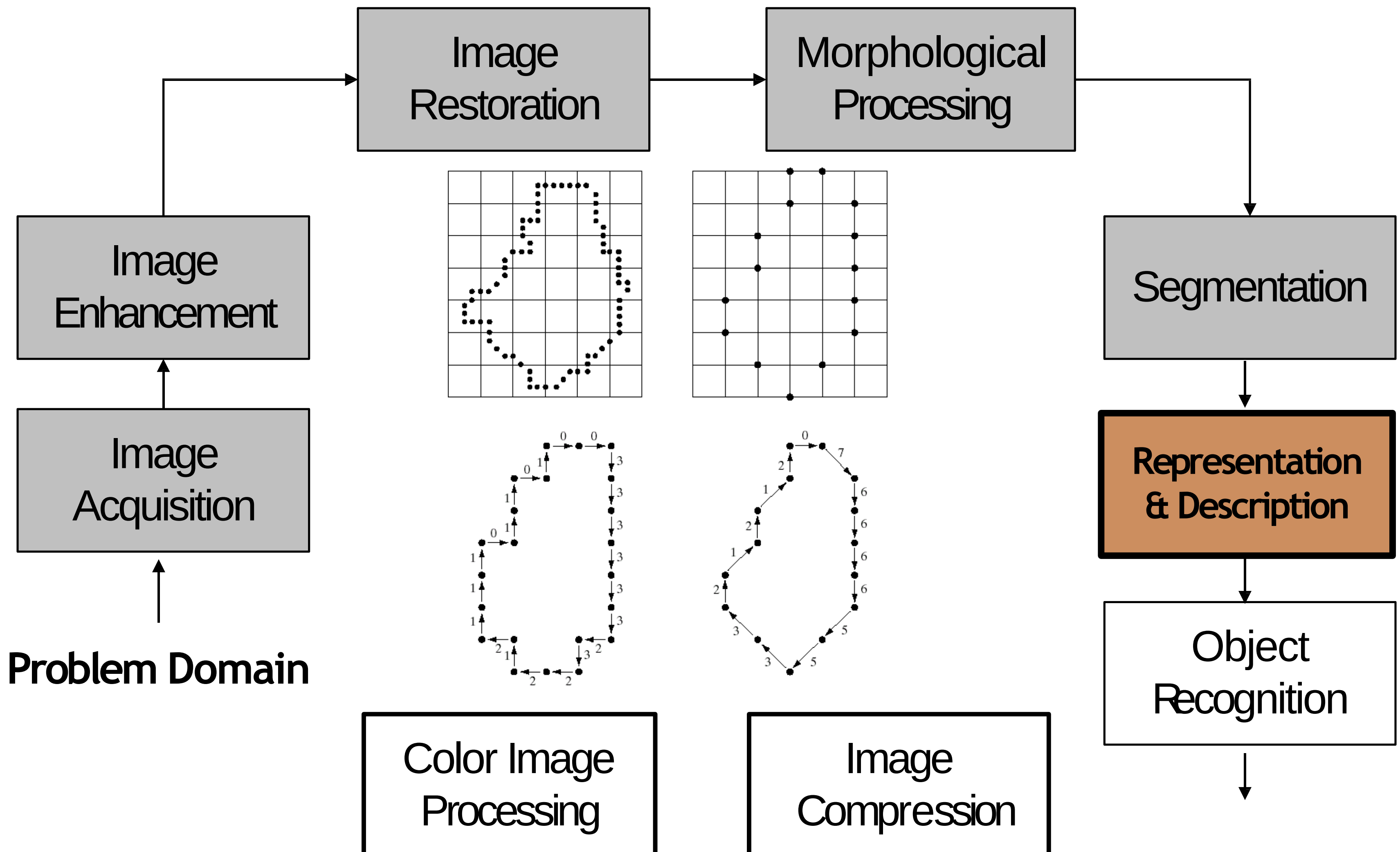
- Image Segmentation algorithms generally are based on one of two basic properties of intensity values:: Discontinuity and Similarity

- Through Discontinuity the approach is to partition an image based on abrupt changes in intensity, such as edges in an image

- Through Similarity the approach is based on partitioning an image into regions that are similar according to a set of predefined criteria. Thresholding, region growing, region splitting and merging are examples of methods in this category



Key Stages in Digital Image Processing



Representation and Description

- A segmented region can be represented by boundary pixels or by internal pixels

- Representing region in 2 ways
  - in terms of its external characteristics (its boundary) focus on shape
  - characteristics
  - in terms of its internal characteristics (its region) focus on regional properties, e.g., color, texture

- sometimes, we may need to use both ways

- The description of a region is based on its representation,
  - for example a boundary can be described by its length

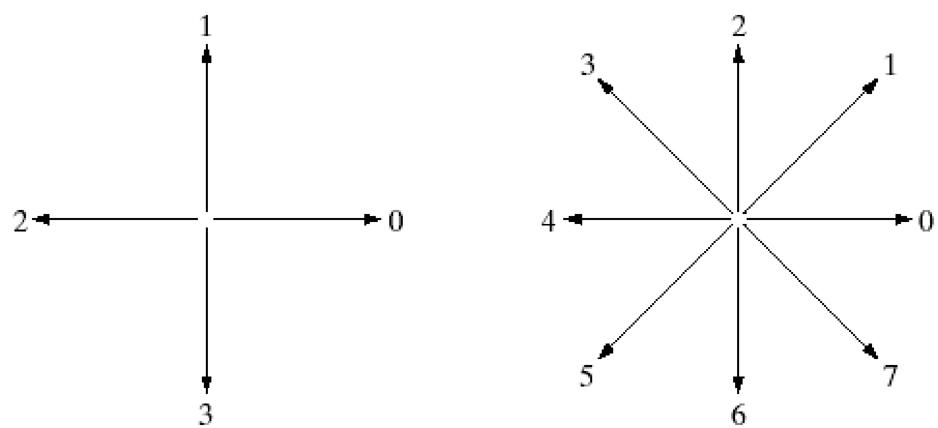
Representation and Description

- Chain Codes are used to represent the boundary by a connected sequence of straight line segments of specified length and direction

- Typically this representation is based on 4 or 8 connectivity of the segments

a b

FIGURE 11.1 Direction numbers for (a) 4-directional chain code, and (b) 8-directional chain code.



Key Stages in Digital Image Processing

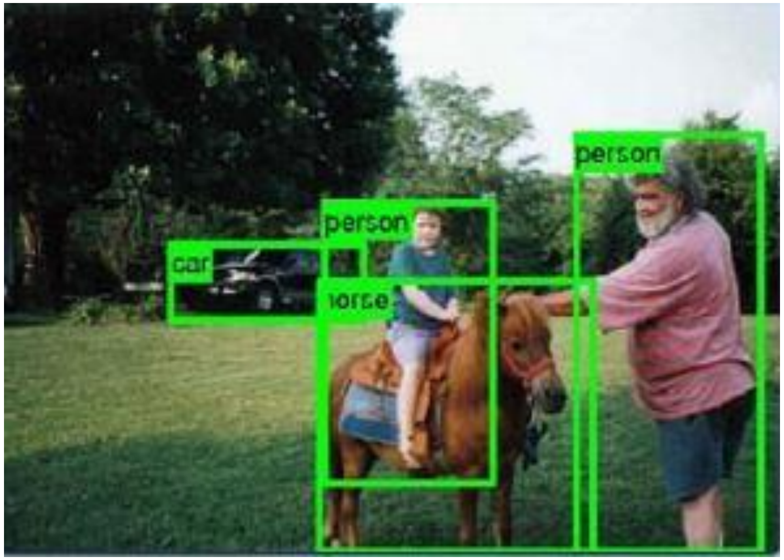
Image Restoration

Morphological Processing

Image Enhancement

Image Acquisition

Problem Domain



Segmentation

Representation & Description

Object Recognition

Color Image Processing

Image Compression

Object Recognition

- Recognition is the process that assigns a label to an object based on its descriptors

- A pattern is an arrangement of descriptors also known as features

- A pattern class is a family of patterns that share some common properties

- Pattern recognition by machine involves techniques for assigning patterns to their respective classes automatically and with as little human intervention as possible

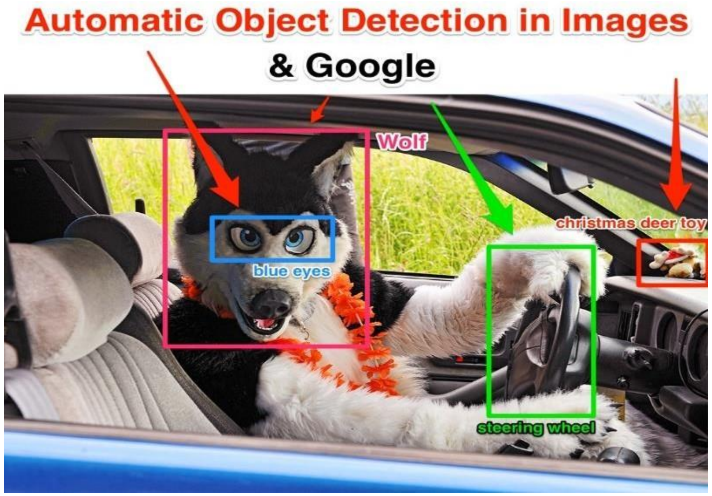


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Image Compression

Image Compression

- Image compression is minimizing the size in bytes of a graphics file without degrading the quality of the image to an unacceptable level.

- The reduction in file size allows more images to be stored in a given amount of disk or memory space. It also reduces the time required for images to be sent over the Internet or downloaded from Web pages.

- Image Compression methods can be based on either:

  - ✓ Lossy Compression methods

  - ✓ Lossless Compression methods

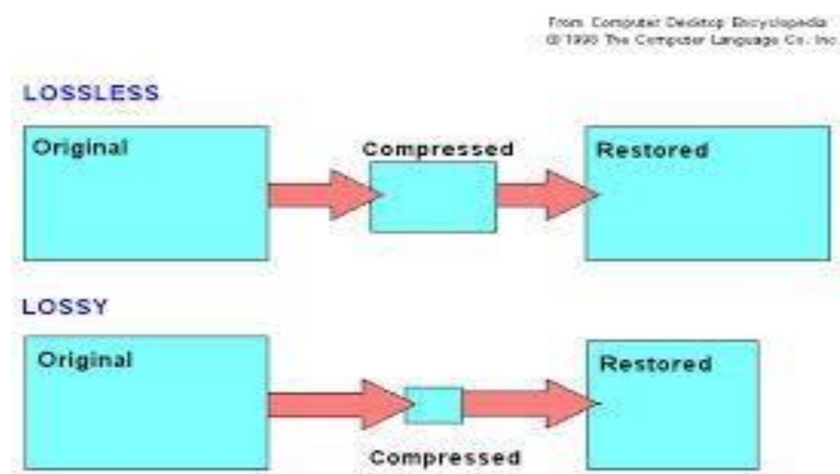

Image Compression

Original Image (lossless PNG, 60.1 KiB size) — uncompressed is 108.5 KiB

Low compression (84% less information than uncompressed PNG, 9.37 KiB)

Medium compression (92% less information than uncompressed PNG, 4.82 KiB)

High compression (98% less information than uncompressed PNG, 1.14 KiB)

Key Stages in Digital Image Processing

Image Restoration

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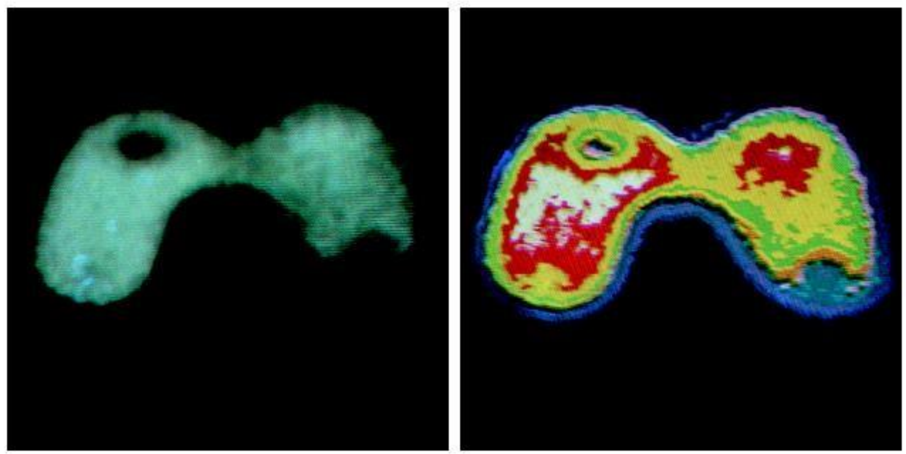
Image Enhancement



Segmentation

Image Acquisition

Representation & Description

Object Recognition

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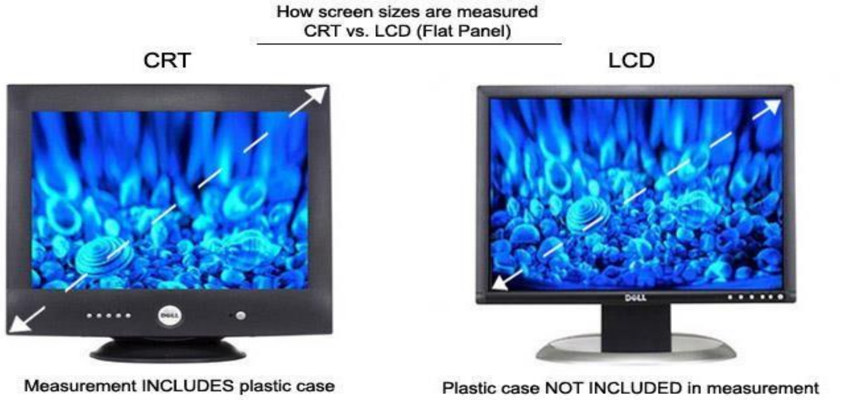
Color Image Processing

Image Compression

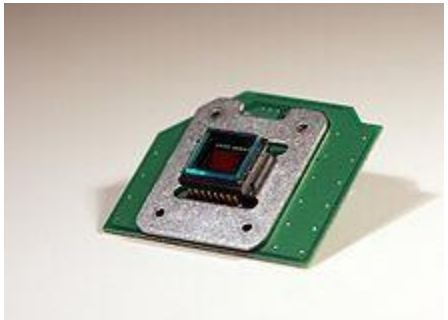
Image Processing Components

- Image Sensing device
- Storage Media
- Processing Systems
- Displays [5]
- Communication Media
- Hardcopy devices (e.g Printer)



Camera

- Lens (CMount, CSMount) [3-4]
- Optical Filter (Selectivity in EM waves)
- Imaging Sensor (CCD Sensor ,CMOS Sensor)[6]
- Pixel count
- Optical vs Digital Zoom



Camera Lens

- The function of the lens in the camera is to direct the light source to the camera
sensor to help focusing the image.

- The main difference of the different lens brands will be the coating that they use.

- Different lens coating will give varying results from sharpness to color reproduction.

- Some "legendary" brands of camera/lens are Carl Zeiss, Leica, Schneider Kreuchnach, etc

Camera Filter/Optical Filter

- Camera filters alter the properties of light entering the camera lens for the purpose of improving the image being recorded.

- The filter can be a square or oblong shape mounted in a holder accessory, or, more commonly, a glass or plastic disk with a metal or plastic ring frame, which can be placed in front of the lens

- Filters can affect contrast, sharpness, color, and light intensity, either individually, or in various
combinations.

- The negative aspects of using filters, though often negligible, include the possibility of loss of image definition if using dirty or scratched filters

Pixel count [6]

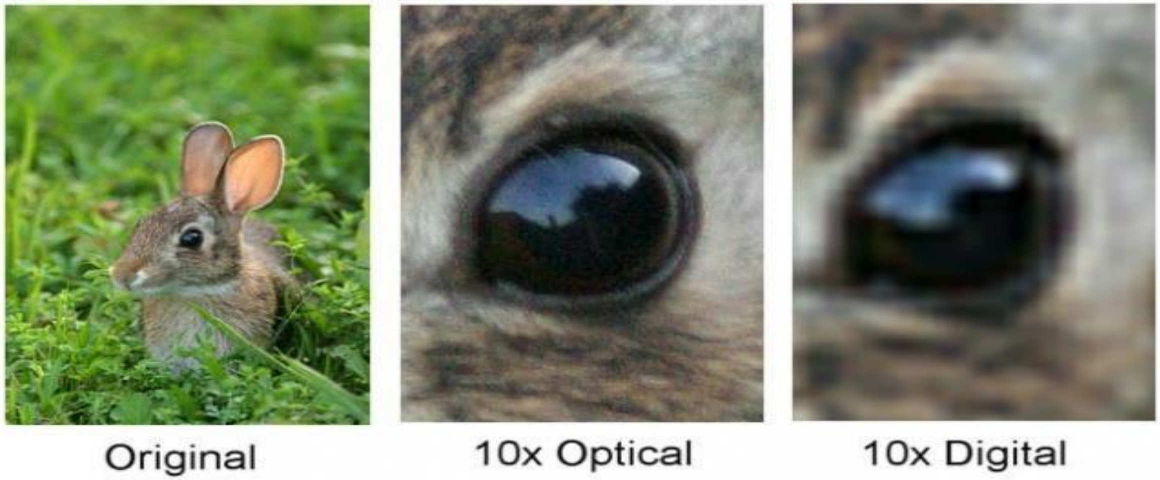
- Gross Pixel count
  - The gross count refers to the total number of pixels on the sensor

- Effective Pixel count
  - Effective count tells you how many pixels will be used when taking video or still photos

- A conventional sensor in, for example, a 12MP (megapixel) camera has an almost equal number of effective pixels (11.9MP). The 0.1% of pixels left over after counting the effective pixels. They are used to determine the edges of an image and to provide color information

Optical vs Digital Zoom [7]

- Optical Zoom
  - Optical zoom is when the lens actually moves in and out and gets you closer to the object. An optical zoom is a “real zoom”.
- Digital Zoom
  - Digital pictures are made up of tons of tiny dots called pixels. A digital zoom just takes those small pixels and enlarges them internally.



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